

CareFlow: Autonomous Healthcare Coordination System

1. Project Essence

CareFlow is an advanced, autonomous healthcare coordination system designed to transition clinical workflows from reactive to proactive. By analyzing 90-day windows of longitudinal patient data, the system identifies high-risk trends before they escalate into critical medical emergencies.

2. The Core Problem & Solution

- **The Problem:** Modern healthcare suffers from “reactive lag.” Worsening lab results or missed appointments often go unnoticed until a patient is readmitted, leading to poor clinical outcomes and increased costs.
 - **The Solution:** An autonomous engine that monitors patient engagement and clinical stability. It generates “Action Proposals” that route through a Human-in-the-Loop (HITL) workflow, ensuring AI-driven insights are always validated by clinical expertise.
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3. Key Technological Pillars

3.1. Proactive Risk Engine

The heart of CareFlow is its custom risk scoring algorithm. It evaluates patients across four weighted dimensions: - **Engagement (1.5x):** Tracking adherence to appointments. - **Clinical (3.0x):** Monitoring volatility in lab results (e.g., Blood Glucose, BP, SpO2). - **Chronic (1.0x):** Accounting for the complexity of existing conditions. - **Instability (0.5x):** Measuring time elapsed since last clinical interaction.

Logic: - **Score ≥ 9.0 :** Immediate Escalation (Emergency consult/Home visit). - **Score ≥ 5.0 :** Follow-up Required (Phone call/Wellness check).

3.2. Role-Based Specialized Portals

- **Doctor Portal:** Focused on clinical decision-making. Doctors review AI-generated proposals, view 90-day trend lines, and approve/reject actions with clinical reasoning.
- **Staff Coordinator Portal:** Focused on operational logistics. Manages patient records, monitors staff/doctor presence, and tracks the execution of clinical decisions.

- **Admin Portal:** Focused on system integrity. Features real-time dashboards, inventory management, user auditing, and the manual trigger for the Risk Engine.

3.3. Immutable Audit Trail

To ensure compliance and accountability, CareFlow implements database-level triggers that log every modification (INSERT, UPDATE, DELETE) into a JSON-based audit log. This provides an unalterable history of how clinical decisions were reached.

4. Technical Architecture

4.1. The Stack

- **Backend:** Python 3.11 with FastAPI (for high-performance, asynchronous I/O).
- **Database:** MySQL 8.0 with SQLAlchemy 2.0 ORM.
- **Frontend:** React 19 powered by Vite, utilizing Tailwind CSS v4 for a modern, fluid UI.
- **Authentication:** JWT-based stateless authentication with role-level guards.

4.2. Database Schema Highlights

- `patients`: Stores identity and chronic conditions.
 - `risk_scores`: Historical record of every patient's calculated risk.
 - `action_proposals`: State-machine tracking pending vs. resolved actions.
 - `audit_logs`: JSON storage for all system mutations.
 - `inventory`: Real-time tracking of 40+ hospital equipment categories.
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5. Scalability & Data Simulation

CareFlow includes a robust data seeding engine capable of generating: - 100+ Patients with diverse risk profiles. - 3,600+ Clinical Records (Appointments and Lab results). - 8 Departments with realistic staffing and inventory levels. This allows for stress-testing the risk engine and demonstrating the system's ability to handle high-density longitudinal data.

6. Implementation Workflow (HITL)

1. Detection: Risk Engine scans database and calculates scores.
2. Proposal: High-risk patients trigger an ActionProposal.

3. Review: Doctor evaluates the proposal alongside the patient's 90-day history.
 4. Decision: Doctor approves or rejects with a comment.
 5. Execution: Staff Coordinator receives the approved decision for operational action.
 6. Audit: The entire cycle is logged in the immutable audit trail.
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7. Project Structure Overview

- careflow/
 - backend/
 - * main.py: API Gateway & Routes
 - * models.py: Schema Definitions
 - * database.py: Connection Management
 - * auth.py: Security & Permissions
 - * seed_data.py: Simulation Engine
 - frontend/
 - * src/
 - components/: Layouts (Doctor/Staff/Admin)
 - pages/: Specialized Dashboards
 - docs/: Technical Manuals